
Problem Statement: Customer Churn Prediction using Backpropagation Neural Network

Objective:

The goal of this project is to **build and evaluate a deep learning model using a Backpropagation Neural Network (BPNN)** to predict whether a customer will churn (i.e., stop using the service) based on their demographic, usage, and account-related attributes.

Description:

Customer churn is a critical issue in the telecom industry, affecting both profitability and customer retention strategies. By accurately predicting churn using customer data, companies can proactively take action to retain high-risk customers.

In this project, we use the **Telco Customer Churn dataset** to train a neural network model with **Keras and TensorFlow**, utilizing **forward and backward propagation** techniques for model learning. The neural network will classify customers as either likely to churn or not.

The workflow includes:

- Data cleaning and encoding
- Feature scaling
- Neural network model building
- Backpropagation with gradient descent using Keras' optimizer
- Model evaluation with accuracy and loss metrics

Dataset:

- **Name:** Telco Customer Churn Dataset
- **Source:** <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

```
import kagglehub

# Download latest version

path = kagglehub.dataset_download("blastchar/telco-customer-churn")

print("Path to dataset files:", path)
```

- **Attributes:**

- Customer demographics (gender, senior citizen, etc.)
- Services used (Internet, Phone, etc.)
- Account information (tenure, MonthlyCharges, etc.)
- Target variable: **Churn** (Yes/No)

Tools & Libraries Used:

- Python
- Pandas, NumPy, Scikit-learn
- TensorFlow + Keras
- Matplotlib (for visualization)